

AcePharm Mobile — Manual QA Test Plan

App: AcePharm (`com.acepharm.app`) · **Build under test:** v0.1.0 · **Platforms:** iOS & Android

This is the pre-release manual verification pass for the AcePharm mobile app. Automated `flutter test` coverage (116 tests) already checks widget behaviour in isolation — this document covers what automated tests can't: real devices, real network conditions, and end-to-end flows across screens.

How to use this document

Run through every section on **one iOS device/simulator and one Android device/emulator** at minimum. Note the specific model/OS version in the sign-off table at the end.

Check a box only when you've personally verified the behaviour — don't check based on reading the code.

For anything that fails, use the bug template at the bottom instead of just noting "fails."

Sections are ordered to match the natural first-run user journey. Later sections assume earlier ones pass (e.g., Progress testing assumes you have real practice attempts on the account).

Test account setup

Use a fresh email (or delete/recreate the account via the app's own Delete Account flow between runs) — testing onboarding and first-attempt telemetry against an already-onboarded account gives false confidence.

1. Authentication

1.1 Sign up

- ☐ Empty form → tapping "Create account" shows both "Enter a valid email address" and "Use at least 8 characters" without calling the network.
- ☐ Password field: typing shows the strength meter live (Weak/Fair/Strong with 1/2/3 filled segments), and it's hidden entirely when the field is empty.
- ☐ Confirm-password mismatch is caught before submission ("Passwords do not match").
- ☐ Successful sign-up lands directly in onboarding Step 1 (no separate "verify email" gate).
- ☐ Signing up with an email that already has an account shows "An account already exists for that email." — not a raw error.

1.2 Log in

- ☐ Empty form → "Enter a valid email address" / "Enter your password", no network call.
- ☐ Wrong password → "Incorrect email or password." (not "wrong-password" or a stack trace).
- ☐ Correct credentials → lands on the dashboard if already onboarded, or onboarding Step 1 if not.
- ☐ "Forgot password?" opens the reset screen; "Sign up" opens the sign-up screen.

1.3 Forgot password

- ☐ Invalid email format is caught client-side before any network call.
- ☐ Valid email → confirmation view ("Reset link sent") replaces the form; the form is not still reachable to resubmit.
- ☐ "Back to login" returns to the login screen.
- ☐ The actual reset email arrives and its link successfully resets the password (test against a real inbox at least once per release).

1.4 Session invalidation

- ☐ With the app signed in, revoke the session from the Firebase console (or wait for a token to genuinely expire) and trigger any API call. The app should fall back to the login screen on its own, not hang on a permanently-failing screen.

2. Onboarding (5 steps)

- ☐ Step 1 (Training stage): all 6 options render (MPharm Year 2/3/4, Foundation Trainee, Oriel, IP); Continue is disabled until one is picked.
- ☐ Step 2 (Primary goal): the 4 presets plus "Other" with a free-text field that must be non-empty before Continue enables.
- ☐ Step 3 (Exam date): optional — Continue works with no date set.
- ☐ Step 4 (Daily goal): stepper respects its min/max bounds; defaults to 20.
- ☐ Step 5 (University): list is populated (29 UK MPharm schools should be present — confirms the university seed reached production); optional, Continue works either way.
- ☐ Back button works from every step except Step 1 (no back arrow shown there).
- ☐ "Finish" on Step 5 submits and lands on the dashboard.
- ☐ Killing the app mid-onboarding and reopening it resumes at onboarding (not skipped), since `hasCompletedOnboarding` wasn't set server-side yet.
- ☐ Submission failure (e.g., airplane mode at the Finish tap) shows an inline error and stays on the wizard rather than losing the collected answers.

3. Dashboard

- ☐ Revision ring shows the real `answered/target` count for today, not a placeholder.
- ☐ Exam countdown ticker appears only when an exam date was set during onboarding, and reads correctly (singular "day" at exactly 1, "today" at 0, no negative count past the date).
- ☐ Weekly Ace Clinical Insight card loads and shows a real paragraph (not stuck loading).
- ☐ Recommendation card shows an explicit reason ("Recommended because...") — never an unexplained suggestion.
- ☐ App bar icons all work: Progress, Subscription, Settings, Logout.
- ☐ Logout returns to the login screen and a subsequent app relaunch does **not** auto sign back in.

4. Session Builder → Practice

4.1 Session Builder

- ☐ All 19 GPhC domains are selected by default; "Clear all" and individual toggles work.
- ☐ Deselecting all domains disables "Start session" with an explanatory message.
- ☐ Switching Learn ↔ Timed Exam mode re-estimates the available question count.
- ☐ "Start session" with zero available questions for the current filter is handled gracefully (no crash, clear message).

4.2 Question screen — core loop

- ☐ Submit is disabled until **both** an option and a confidence rating are chosen (Product Invariant #2).
- ☐ Selecting an option before submitting shows **no** green/red colour or check/cross icon anywhere (Product Invariant #1 — zero pre-submission colour).
- ☐ After submitting, feedback always pairs an icon **and** a text label **and** colour — never colour alone (Product Invariant #3).
- ☐ Every incorrect option shows its own distinct rationale, not a single shared explanation (Product Invariant #4).
- ☐ "Cover options" toggle hides the option list; tapping the hidden-state banner reveals it again; the toggle disables once answered.
- ☐ First-attempt badge is correct: shows "first attempt" language only the very first time a question is answered, and different language on repeat attempts of the same question.
- ☐ Bookmark icon toggles and persists — leave the question and come back, or restart the app, and the bookmark state should hold.
- ☐ The notes editor opens, saves, and reopens with the saved note.

4.3 Ask Ace

- ☐ "Ask Ace about this question" appears only after an explanation is available.

- ☐ A normal clinical question gets a grounded answer with at least one citation shown.
- ☐ A question clearly outside the reviewed curriculum (e.g., ask about an unrelated drug class never covered) gets the graceful refusal — visually distinct (neutral, not styled like a confident answer) and never a fabricated answer.

4.4 Timed Exam mode

- ☐ Countdown timer visibly runs and the mode withholds rationales until the question is answered, matching the "Pearson VUE" exam-condition framing.

5. Offline resilience (Product Invariant #6)

This is the newest and highest-risk area — test it thoroughly on a real device, not just a simulator (simulators can behave oddly around airplane mode).

- ☐ Start a question, enable **airplane mode**, then submit. Expect: no hard error — a neutral "Saved — you're offline. This will be graded once you're back online." message, and the footer offers "Next question" (not stuck on "Submit answer").
- ☐ While still offline, answer 2–3 more questions the same way.
- ☐ Return to the dashboard while still offline — a banner reading "N answers will sync once you're back online" should show the correct count.
- ☐ Re-enable connectivity. Within a few seconds (no manual action required), the banner count should drop to zero as the queued answers sync in the background.
- ☐ Force-quit the app while answers are still queued offline, relaunch with connectivity restored, and confirm the queue still flushes automatically (proves the queue survives an app restart, not just backgrounding).
- ☐ A **real** server error (not connectivity — e.g., trigger a 500 if you have a way to) still shows the old inline "Couldn't submit your answer. Try again." error, not the offline notice. This confirms the two failure paths aren't conflated.

6. Progress & mastery grid

Needs an account with a real mix of correct/incorrect attempts across several subtopics — either use a seasoned test account or generate attempts first.

- ☐ All 5 pillars render as visibly distinct numbers (First-Attempt, Practice, Repeat, Calibration, Coverage) — confirm none of them are silently equal/merged by coincidence of test data; if they're suspiciously identical, generate more varied attempts and recheck.
- ☐ Confidence Calibration grid shows Overconfident / Calibrated / Underconfident correctly given the account's actual confidence-vs-correctness pattern.
- ☐ 19-topic mastery grid shows all expected status labels somewhere across the categories: Not started, First pass, Needs attention, Developing, Secure. "Due for review" requires a subtopic that reached Secure and then enough time has passed per its last attempt's freshness window (1/3/7/21 days depending on correctness+confidence) — this one may need a seeded/backdated test account to observe directly within a QA cycle; if so, verify via the [progress-calculator.test.ts](#) backend tests instead and note that here.

7. Subscription

- ☐ Free/Explorer account: shows "Explorer (Free)" and explanatory copy; no "Manage billing" button (there's no Stripe customer to manage).
- ☐ Paid account: shows the correct plan name, renewal date, and a working "Manage billing" button that opens the real Stripe customer portal in the browser/webview.
- ☐ A subscription set to cancel at period end shows the "not renewing" warning with the correct access-until date.
- ☐ Portal load failure (e.g., airplane mode) shows an inline error, not a crash.

8. Account deletion

Use a disposable test account for this section — it is irreversible.

- ☐ Settings → Account → Delete account opens the two-step flow.
- ☐ Step 1 clearly lists what's lost (practice history, notes, bookmarks, subscription) and has a working Cancel that backs out with zero side effects.
- ☐ Step 2 requires the account password; an empty submission is caught client-side; a wrong password shows an error and does **not** delete anything.
- ☐ Correct password → the account is actually gone: attempting to log back in with the same credentials fails, and the app lands on the login screen automatically (no manual navigation needed).
- ☐ If the account had an active paid subscription, confirm in the Stripe dashboard that the subscription was cancelled, not left billing a deleted account.

9. Cross-cutting checks

- ☐ **Touch targets:** spot-check a few option rows and icon buttons with a ruler/accessibility inspector — should be $\geq 44 \times 44$ pt.
- ☐ **Accessibility:** turn on VoiceOver (iOS) or TalkBack (Android) and swipe through the dashboard revision ring, a question screen, and the mastery grid — content should announce sensibly, not just raw numbers with no context.
- ☐ **Crash reporting:** trigger a deliberate crash in a debug/staging build (e.g., a temporary `throw` behind a hidden button) and confirm it appears in Firebase Crashlytics within a few minutes. Remove the trigger before release.
- ☐ **Analytics:** check Firebase Analytics DebugView while manually running through onboarding and a practice session — confirm `onboarding_step_viewed`, `onboarding_completed`, `session_created`, and `question_answered` events all fire with sensible parameters.
- ☐ **Cold start:** force-quit and relaunch at every major screen (dashboard, mid-question, mid-onboarding) — nothing should crash or lose critical state unexpectedly.
- ☐ **Rotation / multitasking** (tablet or split-screen where applicable): no layout overflow errors on the question screen or progress grid.

Bug report template

```

**Screen/Flow:**
**Device/OS:**
**Steps to reproduce:**
1.
2.
**Expected:**
**Actual:**
**Screenshot/recording:**
**Severity:** Blocker / Major / Minor / Cosmetic

```

Sign-off

Platform	Device/OS version	Tester	Date	Result
iOS				☐ Pass ☐ Pass with notes ☐ Fail
Android				☐ Pass ☐ Pass with notes ☐ Fail